

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

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**Device Specifications:**

Device Type: Matrix Switcher  
Manufacturer: Broaddata Communications  
Firmware Version: N/A  
Model(s): LBS-MSW88

**Tested on the Following Software and Firmware Versions:**

| IP Link Pro Control Processor<br>Firmware | Touch Link Panel Firmware | GS Version |
|-------------------------------------------|---------------------------|------------|
| 2.04.0000-b007                            | 2.01.0001-b001            | 1.2.2      |

**Version History**

| Module Version | Date      | Notes           |
|----------------|-----------|-----------------|
| 1_0_1_0        | 1/23/2017 | Initial Version |

**Module Notes**

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:  
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:  
InterfaceName.connectionCounter = 5

**Control Commands, States, and Qualifiers:**

|                              |                                         |                                                       |
|------------------------------|-----------------------------------------|-------------------------------------------------------|
| <b>APMControl</b>            | 'On'                                    | 'Off'                                                 |
| <b>AudioMatrixTieCommand</b> | None                                    |                                                       |
| <b>['Input']</b>             | '1' – '8'                               |                                                       |
| <b>['Output']</b>            | '1' – '8'                               |                                                       |
| <b>['Type']</b>              | 'Stereo'<br>'Select Video Input Stereo' | 'Extract'<br>'Select Video Extract'      'Output ARC' |
| <b>IRControl</b>             | 'On'                                    | 'Off'                                                 |
| <b>IRMatrixTieCommand</b>    | None                                    |                                                       |
| <b>['Input']</b>             | '1' – '8'                               |                                                       |
| <b>['Output']</b>            | '1' – '8'                               | 'All'                                                 |
| <b>KeyControl</b>            | 'On'                                    | 'Off'                                                 |
| <b>OutputControl</b>         | 'On'                                    | 'Off'                                                 |
| <b>['Output']</b>            | '1' – '8'                               | 'All'                                                 |
| <b>Power</b>                 | 'On'                                    | 'Off'                                                 |
| <b>VideoMatrixTieCommand</b> | None                                    |                                                       |
| <b>['Input']</b>             | '1' – '8'                               |                                                       |
| <b>['Output']</b>            | '1' – '8'                               | 'All'                                                 |

**Status Available:**

|                             |             |                |
|-----------------------------|-------------|----------------|
| <b>ConnectionStatus</b>     | 'Connected' | 'Disconnected' |
| <b>IRControl</b>            | 'On'        | 'Off'          |
| <b>IRInputTieStatus</b>     | 'Tied'      | 'Untied'       |
| ['Input']                   | '1' – '8'   |                |
| ['Output']                  | '1' – '8'   |                |
| <b>IROutputTieStatus</b>    | '1' – '8'   |                |
| ['Output']                  | '1' – '8'   |                |
| <b>KeyControl</b>           | 'On'        | 'Off'          |
| <b>OutputControl</b>        | 'On'        | 'Off'          |
| ['Output']                  | '1' – '8'   | 'All'          |
| <b>Power</b>                | 'On'        | 'Off'          |
| <b>VideoInputTieStatus</b>  | 'Tied'      | 'Untied'       |
| ['Input']                   | '1' – '8'   |                |
| ['Output']                  | '1' – '8'   |                |
| <b>VideoOutputTieStatus</b> | '1' – '8'   |                |
| ['Output']                  | '1' – '8'   |                |

***Cable and Adapter Requirements:***

Captive Screw to Female DB9 RS-232 Serial Cable

***Notes for the Device:***

***Serial communication:***

**Port Type:** RS-232

**Baud Rate:** 57600

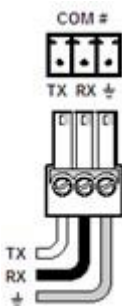
**Data Bits:** 8

**Parity:** None

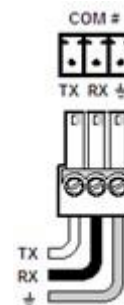
**Stop Bits:** One

**Flow Control:** None

***Pin Assignments Diagram***



| Signal | Main Cable | Signal |
|--------|------------|--------|
| TxD    |            | TxD    |
| RxD    |            | RxD    |
| GND    |            | GND    |



***General Notes:***

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***Network communication:***

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

|                                       |              |
|---------------------------------------|--------------|
| <b>Port Type:</b>                     | Ethernet     |
| <b>Default Port:</b>                  | 23           |
| <b>Logon Credentials Supported:</b>   | No           |
| <b>Multi-Connection Capabilities:</b> | Undetermined |
| <b>Port Changeability:</b>            | No           |

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***Ethernet Module Configuration Description:***

Please refer to user manual for settings and changes to the network communication

***Notes for the Device:***

**Appendix A. Set Commands**

|                                                                     |                              |
|---------------------------------------------------------------------|------------------------------|
| APM Control Off                                                     | APM OFF\x0D\x0A              |
| APM Control On                                                      | APM ON\x0D\x0A               |
| Audio Matrix Tie Command Input 1 Output 1 Type Stereo               | AUD TX 01 ANA 01\x0D\x0A     |
| Audio Matrix Tie Command Input 8 Output 8 Type Select Video Extract | AUD TX 08 FOEXT\x0D\x0A      |
| IR Control Off                                                      | IR OFF\x0D\x0A               |
| IR Control On                                                       | IR ON\x0D\x0A                |
| IR Matrix Tie Command Input 1 Output 1                              | MXIR 01 FR 01 Output\x0D\x0A |
| IR Matrix Tie Command Input 8 Output All                            | MXIR 00 FR 08 Output\x0D\x0A |
| Key Control Off                                                     | KEY OFF\x0D\x0A              |
| Key Control On                                                      | KEY ON\x0D\x0A               |
| Output Control Off Output 1                                         | OUT 01 OFF\x0D\x0A           |
| Output Control Off Output 2                                         | OUT 02 OFF\x0D\x0A           |
| Output Control Off Output 3                                         | OUT 03 OFF\x0D\x0A           |
| Output Control Off Output 4                                         | OUT 04 OFF\x0D\x0A           |
| Output Control Off Output 5                                         | OUT 05 OFF\x0D\x0A           |
| Output Control Off Output 6                                         | OUT 06 OFF\x0D\x0A           |
| Output Control Off Output 7                                         | OUT 07 OFF\x0D\x0A           |
| Output Control Off Output 8                                         | OUT 08 OFF\x0D\x0A           |
| Output Control Off Output All                                       | OUT 00 OFF\x0D\x0A           |
| Output Control On Output 1                                          | OUT 01 ON\x0D\x0A            |
| Output Control On Output 2                                          | OUT 02 ON\x0D\x0A            |
| Output Control On Output 3                                          | OUT 03 ON\x0D\x0A            |
| Output Control On Output 4                                          | OUT 04 ON\x0D\x0A            |
| Output Control On Output 5                                          | OUT 05 ON\x0D\x0A            |
| Output Control On Output 6                                          | OUT 06 ON\x0D\x0A            |
| Output Control On Output 7                                          | OUT 07 ON\x0D\x0A            |
| Output Control On Output 8                                          | OUT 08 ON\x0D\x0A            |
| Output Control On Output All                                        | OUT 00 ON\x0D\x0A            |
| Power Off                                                           | POFF\x0D\x0A                 |
| Power On                                                            | PON\x0D\x0A                  |
| Video Matrix Tie Command Input 1 Output 1                           | OUT 01 FR 01\x0D\x0A         |
| Video Matrix Tie Command Input 8 Output All                         | OUT 00 FR 08\x0D\x0A         |

**Appendix B. Query Commands**

|                                      |               |
|--------------------------------------|---------------|
| IR Input Tie Status Input 1 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 1 Output 8 | MXSTA\x0D\x0A |

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|                                      |               |
|--------------------------------------|---------------|
| IR Input Tie Status Input 2 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 2 Output 8 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 3 Output 8 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 4 Output 8 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 5 Output 8 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 6 Output 8 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 1 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 2 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 3 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 4 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 5 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 6 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 7 | MXSTA\x0D\x0A |
| IR Input Tie Status Input 7 Output 8 | MXSTA\x0D\x0A |

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|                                      |                |
|--------------------------------------|----------------|
| IR Input Tie Status Input 8 Output 1 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 2 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 3 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 4 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 5 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 6 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 7 | MXSTA\x0D\x0A  |
| IR Input Tie Status Input 8 Output 8 | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 1        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 2        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 3        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 4        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 5        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 6        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 7        | MXSTA\x0D\x0A  |
| IR Output Tie Status Output 8        | MXSTA\x0D\x0A  |
| IR Control                           | STATUS\x0D\x0A |
| Key Control                          | STATUS\x0D\x0A |
| Output Control                       | STATUS\x0D\x0A |
| Power                                | STATUS\x0D\x0A |
| Video Input Tie Status               | STATUS\x0D\x0A |